Q1.

An alloy is a mixture of elements.
The table shows the mass of each element present in 100 g of five different alloys, **bronze, solder, steel, stainless steel** and **brass.**

alloy	mass of each element in 100 g of alloy							
	lead (g)	**tin (g)**	**copper (g)**	**zinc (g)**	**carbon (g)**	**iron (g)**	**chromium (g)**	**nickel (g)**
bronze		4	95	1				
solder	62	38						
steel					1	99		
stainless steel						70	20	10
brass			67	33				

(a) Which **alloy** in the table above contains an element which is a non-metal?

.....

1 mark

(b) Which **two alloys** in the table contain only **two metals**?

..... and .....

1 mark

(c) Another alloy called nichrome contains only the elements chromium and nickel.
100 g of nichrome contains 20 g of chromium.

How much nickel does it contain?

…… g

1 mark

(d) Before 1992, two-pence coins were made of bronze.
Steel rusts but bronze does **not** rust.

(i) Why does bronze **not** rust?
Use information in the table above to help you.

.....

.....

1 mark

(ii) Rusting requires water and a gas from the air.
Give the name of this gas.

.....

1 mark
maximum 5 marks

Q1.

(a) steel

*do **not** accept 'stainless steel'*
*do **not** accept 'carbon'*

1 (L5)

(b) • brass

• solder

answers may be in either order
both answers are required for the mark

1 (L5)

(c) 80

accept '100 – 20'

1 (L5)

(d) (i) it does not contain iron

accept 'it does not contain steel'
*accept 'only iron rusts' **or** 'only steel rusts'*
accept 'it is made of tin, copper and zinc'

1 (L6)

(ii) oxygen

accept 'O_2'

1 (L6)

[5]